

A facile strategy to improve the electrochemical stability of a lithium ion conducting $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ solid electrolyte



Yan Sun^{a,b}, Wenning Yan^{a,b}, Li An^c, Bingbin Wu^c, Kaifu Zhong^c, Ruizhi Yang^{a,b,*}

^a Soochow Institute for Energy and Materials Innovations, College of Physics, Optoelectronics and Energy & Collaborative Innovation Center of Suzhou Nano Science and Technology, Soochow University, Suzhou, Jiangsu 215006, China

^b Institute of Chemical Power Sources, Suzhou, Jiangsu 215006, China

^c Contemporary Amperex Technology Limited, Ningde, Fujian 352000, China

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ABSTRACT

A super solid electrolyte of Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ is designed and fabricated to achieve high electrochemical stability. Among the Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ solid electrolytes, the composition of $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ exhibits the highest ionic conductivity of $7.04 \times 10^{-4} \text{ S cm}^{-1}$ with an activation energy of $28.27 \text{ kJ mol}^{-1}$ at 25°C . The electrochemical window of $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ is between 0 and 6 V (vs. Li), which is higher than the typical 5 V window of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$. As compared with $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ demonstrates a significantly reduced polarization and a better cycle stability though its ionic conductivity is slightly lower than that of the former. The electrochemical stability of Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ can be attributed to the structural stability resulting from the stable interaction between Ba^{2+} and S^{2-} in the structure.

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1. Introduction

To solve the safety issues associated with the flammable organic liquid electrolyte in lithium-ion (Li-ion) batteries, solid electrolyte with high ionic conductivity is highly desired. Sulfide-based solid electrolytes, such as $\text{Li}_2\text{S-SiS}_2$ [1], $\text{Li}_2\text{S-SiS}_2\text{-Li}_x\text{MO}_y$ ($\text{M} = \text{B, Al, Si, S, Ga}$) [2–4], $\text{Li}_2\text{S-P}_2\text{S}_5$ [5,6] and $\text{Li}_{10 \pm 1}\text{MP}_2\text{S}_{12}$ ($\text{M} = \text{Ge, Si, Sn, Al or P}$), have attracted much attention because of their superior ionic conductivity, inflammability and wide electrochemical window [1–4]. Among the sulfide-based solid electrolytes, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS) has achieved the highest bulk ionic conductivity of $12 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature (25°C) [7,8]. However, the first principles study indicated that the LGPS is a metastable phase and it is not stable against reduction by Li at low voltage or extraction of Li at high voltage [9]. Furthermore, the high symmetry structure and high density of vacancies were found to make the structure of LGPS less stable as revealed by the molecular dynamic simulations [10]. It's obvious that the stability of LGPS is a major limitation for the application of this material in the solid-state Li-ion batteries.

The three-dimensional (3D) framework of LGPS consists of MS_4 tetrahedra ($\text{M} = \text{Ge, P}$), LiS_6 octahedra and LiS_4 tetrahedra, where the MS_4

tetrahedra are linked by the LiS_6 octahedra, whose corners are shared with the MS_4 tetrahedra. According to the first principle calculations reported by Hu and Ouyang et al. [11], both the interionic coulomb interaction and Van Der Waals interaction contribute to the structural stability of LGPS. Therefore, if the lithium ions in framework are partially substituted by divalent or trivalent cations, the coulomb interaction between S ions and divalent/trivalent cations will be enhanced as compared to that between S ions and Li ions. This will lower the total energy of LGPS framework and enhance its structural stability.

In this work, the divalent cation Ba^{2+} was selected as the substitutive element for Li in LGPS. Effects of the doping amount of Ba^{2+} on the structure, ionic conductivity and electrochemical stability were studied. The solid electrolyte with the composition of $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ shows a lower polarization and a better cycle stability than non-doped LGPS. To the best of our knowledge, this is the first time that partial Li sites are replaced by divalent metal ions to achieve high structural stability of LGPS.

2. Experimental

2.1. Synthesis of Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$

The materials were prepared by a solid state method. Reagent-grade Li_2S (Alfa-Aesa, 99.9%), P_2S_5 (Aladdin, 99%), GeS_2 (BT Reagent, 99.999%) and BaS (Alfa-Aesa, 99.7%) were used as the precursors. All the synthesis procedures were carried out in an Ar-filled glove box in order to prevent

* Corresponding author at: Soochow Institute for Energy and Materials Innovations, College of Physics, Optoelectronics and Energy & Collaborative Innovation Center of Suzhou Nano Science and Technology, Soochow University, Suzhou, Jiangsu 215006, China.

E-mail address: yangrz@suda.edu.cn (R. Yang).

the materials from reacting with the moisture and oxygen. For the preparation of Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ ($\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$, $x = 0$ to 0.5), the molar ratio of Ge:P:S was fixed to be 1:2:12, while the molar ratio of Li/Ba was controlled to be 9.8:0.1, 9.6:0.2, 9.4:0.3, 9.2:0.4, and 9.0:0.5, respectively. The precursors with different Li/Ba molar ratios were weighed and mixed in zirconia pots followed by the mechanical milling for 24 h using ball milling apparatus (CMT, T1-100). Afterwards, the obtained powder was pressed into pellets under 320 MPa. Then the pellets were carefully sealed into the quartz tube and annealed at 550 °C under Ar for 8 h. The final products were naturally cooled down to room temperature. The as-obtained materials with Li/Ba molar ratio of 9.8:0.1, 9.6:0.2, 9.4:0.3, 9.2:0.4, and 9.0:0.5 are named as $\text{Li}_{9.8}\text{Ba}_{0.1}\text{GeP}_2\text{S}_{12}$ (abbreviated as LBGPS-1), $\text{Li}_{9.6}\text{Ba}_{0.2}\text{GeP}_2\text{S}_{12}$ (abbreviated as LBGPS-2), $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ (abbreviated as LBGPS-3), $\text{Li}_{9.2}\text{Ba}_{0.4}\text{GeP}_2\text{S}_{12}$ (abbreviated as LBGPS-4) and $\text{Li}_{9.0}\text{Ba}_{0.5}\text{GeP}_2\text{S}_{12}$ (abbreviated as LBGPS-5), respectively. For comparison, $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (abbreviated as LGPS) was also synthesized with the same procedure without adding BaS [7].

2.2. Physical characterizations

The X-ray diffraction (XRD, $\text{CuK}\alpha$) measurements were performed on an X'pert PRO diffractometer (PANalytical) with $\text{Cu K}\alpha$ radiation, the diffraction angle ranging from 5° to 80° with a step of 0.02°. The samples were covered by Kapton films to avoid moisture uptake.

Raman spectroscopy of the sample was performed on a Jobin Yvon LabRAM HR 800 instrument with a 633 nm excitation laser at a power of around 1 mW. The samples were sealed into capillary tubes to avoid contacting with moisture.

2.3. Electrochemical measurements

The lithium ionic conductivity was measured in cooling scans from 80 °C to −25 °C by the electrochemical impedance spectroscopy (EIS) using an impedance analyzer (Solartron SI-1470E). Firstly, the samples were cold pressed into dense pellets (thickness: 0.71 mm, diameter: 10 mm) under 320 MPa, and then the carbon coated Al foils, serving as the blocking electrode, were pressed on both sides of the pellets and assembled in a specially designed air-tight cell in an argon-filled glove box. The EIS measurement was performed on this symmetric Al/solid electrolyte (SE) pellet/Al cell by applying 15 mV in the frequency range of 1 MHz to 10 Hz. The ionic conductivity was calculated through the formula $\sigma = L / (A \cdot R)$, where L is the thickness of the pellet, A is the area of the blocking electrode and R is the resistance derived from the impedance spectra. To obtain the conduction activation energy, the impedance data under different temperatures were linearly fitted, and the linearization parameters are listed in Table 1.

The lithium transference number was determined by the chrono-amperometric analysis on a symmetric Li/SE pellet/Li cell with an applied potential of 10 mV. EIS measurement was carried out before and after the chrono-amperometric measurement according to the strategy provided by Bruce and Vincent [12] in the frequency range of 1 MHz to 10 Hz.

Cyclic voltammetry (CV) was carried out on using a potentiostat (Solartron SI-1470E). A stainless steel was used as the counter electrode

and lithium foil acted as the reference electrode. The potential sweep was carried out between −0.5 and 6 V with a scanning rate of 1 mV s^{-1} .

To investigate the electrochemical performance of the samples, the galvanostatic polarization test was carried out on a symmetric Li/SE pellet/Li cell with a current of 0.1 mA using a Land CT2001A battery test system (LANHE, Inc.)

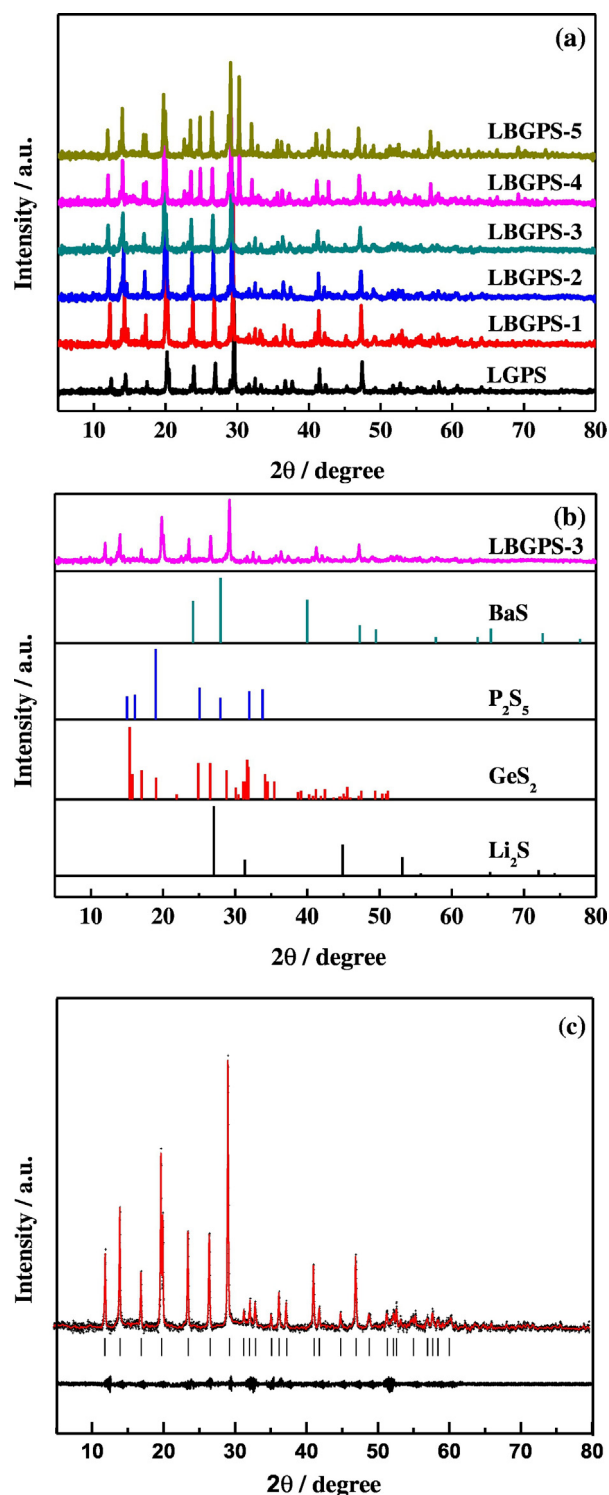


Fig. 1. XRD patterns of (a) Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ with different molar ratios of Li/Ba and (b) LBGPS-3 and the precursors. Rietveld refined XRD pattern of LBGPS-3 (c), unit cell: tetragonal $P4_2/nmc$, $R_{wp} = 9.72\%$, $R_B = 9.94\%$, $\text{GoF} = 9.81$.

Table 1

Parameters determined from the liner fit of the experimental Arrhenius plot. A and B are the slop and intercept of the liner fit plot of equation $y = Ax + B$, respectively. R is the confidence coefficient of the linearization. SD is the standard deviation and N is the number of points in the linearization.

Sample	A	B	R	SD	N
LGPS	−2.77	3.05	−0.999	0.062	6
LBGPS-3	−3.40	4.20	−0.997	0.129	6

Table 2
Crystal structure parameters of LGPS and Ba-substituted LGPS.

Sample	Space group	a/Å	c/Å
LGPS	Tetragonal $P4_2/nmc$	8.694	12.599
LBGPS-1		8.716	12.728
LBGPS-2		8.727	12.908
LBGPS-3		8.734	13.005
LBGPS-4		8.752	13.011
LBGPS-5		8.770	13.053

3. Results and discussion

Fig. 1 shows the XRD patterns of LGPS and $\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$ ($x = 0$ to 0.5). As can be seen in Fig. 1a, all the peaks of $\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$ ($x = 0$ to 0.5) can be indexed with a space group of tetragonal $P4_2/nmc$, coincide with that of LGPS reported by Hassoun [13]. For the samples with low doping amount, especially for LBGPS-1, LBGPS-2 and LBGPS-3, no impurity phases have been observed. While for LBGPS-4 and LBGPS-5, the extra peaks at 24.8° , 30.3° , 42.7° are found, which might be attributed to Ba_2GeS_4 (JCPDS No. 22-0082) and GeS_2 (JCPDS No. 40-0443). This reveals the formation of impurities with high doping amount of Ba in LGPS. The determined crystal structure parameters for LGPS and $\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$ ($x = 0$ to 0.5) are summarized in Table 2. With the increasing of doping amount of Ba, the lattice parameter a , c increase continuously, indicating the successful doping of Ba in LGPS due to the larger Ba^{2+} occupies the Li^+ or Ge^{4+} site in the structure. The phase formed with different compositions (i.e. $\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$, $x = 0$ to 0.5) is checked for those of starting materials in order to examine the existence of impurities. Specifically, the diffraction reflection of a typical Ba-substituted LGPS (i.e. LBGPS-3) is compared with that of Li_2S (JCPDS No. 26-1188), BaS (JCPDS No. 75-0896), GeS_2 (JCPDS No. 40-0443), and P_2S_5 (JCPDS No. 50-0813) as shown in Fig. 1 b. The phases of the raw materials can't be observed in LBGPS-3 and a new phase is formed in LBGPS-3. The Rietveld refinement of the XRD pattern of LBGPS-3 (Fig. 1 c) indicates a contaminant amount lower than 3% in the sample and cell parameters of $a = 8.734 \pm 0.002$ Å and $c = 13.005 \pm 0.003$ Å (Unit cell: tetragonal $P4_2/nmc$).

The Raman spectra of LBGPS with different doping contents of Ba are shown in Fig. 2a. A blue shift of the dominant peaks in LBGPS can be observed with the increasing of Ba content. The spectrum of LBGPS-3 is compared with that of Li_2S , BaS , P_2S_5 and GeS_2 precursors as shown in Fig. 2 b. All the measurements were carried out under the same condition with the wavenumber ranging from 200 to 750 cm^{-1} . The peak at around 348 cm^{-1} can be related to the symmetric stretching mode of the tetrahedral $(\text{GeS}_{0.5})_4$ [13]. While the peak at 365 cm^{-1} is attributed to the Li^+-S^- stretching vibration according to Martin's report [14]. The peak at about 383 cm^{-1} is the symmetry vibration of $(\text{GeS}_{0.5}\text{S}_3)^{3-}$

[14]. For the peak at 414 cm^{-1} , the stretching vibration is associated with $(\text{P}_2\text{S}_7)^{4-}$ [15]. In turn, the peak appeared at 425 cm^{-1} is assigned to the stretching vibration of $(\text{PS}_4)^{3-}$ [16]. Compared with the Raman spectra of LGPS reported by Hassoun [13], there are blue shifts of $(\text{GeS}_{0.5})_4$, $(\text{P}_2\text{S}_7)^{4-}$ and $(\text{PS}_4)^{3-}$ vibration peaks resulted from Ba doping, indicating the binding energy of $(\text{GeS}_{0.5})_4$, $(\text{P}_2\text{S}_7)^{4-}$ and $(\text{PS}_4)^{3-}$ are increased. This reveals that doping of Ba enhances the structural stability of LGPS. The vibration peaks of precursors aren't found in LBGPS-3, which suggests that Ba^{2+} interacts with Ge^{4+} , P^{5+} or S^{2-} in LBGPS. Though Ba^{2+} vibration in the aimed product can't be observed clearly, the peak shift toward high wavenumber with the increasing of doping amount of Ba demonstrates the impact of Ba^{2+} on the structure.

The Nyquist plots of LGPS and LBGPS with different doping content of Ba are shown in Fig. 3a. The corresponding ionic conductivity of these samples at room temperature is also included in Fig. 3a. Note that the ionic conductivity value is only associated with the electrolyte due to the existence of block electrode. LBGPS-3 has the highest ionic conductivity of $7.07 \times 10^{-4}\text{ S cm}^{-1}$ among the Ba-substituted samples. The lower ionic conductivity of the five Ba-substituted samples than that of LGPS can be attributed to the blocking of the lithium conduction pathway by the doping of Ba^{2+} , which indicates that Ba^{2+} most likely occupies Li site. Note that the change of lattice parameters may also change the ionic conductivity. The ionic conductivity of Ba-substituted samples is firstly increased and then decreased sharply with the increasing of doping amount of Ba^{2+} . The reason for the increasing of ionic conductivity at the early stages can be ascribed to the improved structural stability in ab plane due to the interaction between Ba^{2+} and S^{2-} , which may promote the mobility of Li^+ in ab plane and favor the forming of proper tunnel for increasing the Li^+ transference. As for the decreasing of ionic conductivity in LBGPS-4 and LBGPS-5, the formation of miscellaneous phases, including GeS_2 and Ba_2GeS_4 , might not be suitable for the transference of Li^+ . Besides, the Li defects in LBGPS-4 and LBGPS-5 are higher than those in LBGPS-3, suggesting that excessive doping might damage the Li^+ migration tunnel. Among the doping samples, the LBGPS-3 with the highest ionic conductivity was selected for the following electrochemical measurements. The Arrhenius plots of LGPS and LBGPS-3 are displayed in Fig. 3 b. The activation energy of the products is calculated using the equation:

$$\sigma_T = \sigma_0 \exp(-E_a/RT) \quad (1)$$

where σ_T is the ionic conductivity of the synthesized material, σ_0 is the pre-exponential parameter, T is the absolute temperature, E_a is the activation energy, R is the gas constant. The calculated activation energies for LGPS and LBGPS-3 from the Arrhenius plots are 23.03 kJ mol^{-1} and 28.27 kJ mol^{-1} , respectively. The E_a value of LGPS is similar to the value of 24 kJ mol^{-1} reported by Kanno et al. [7]. The Ba-substituted

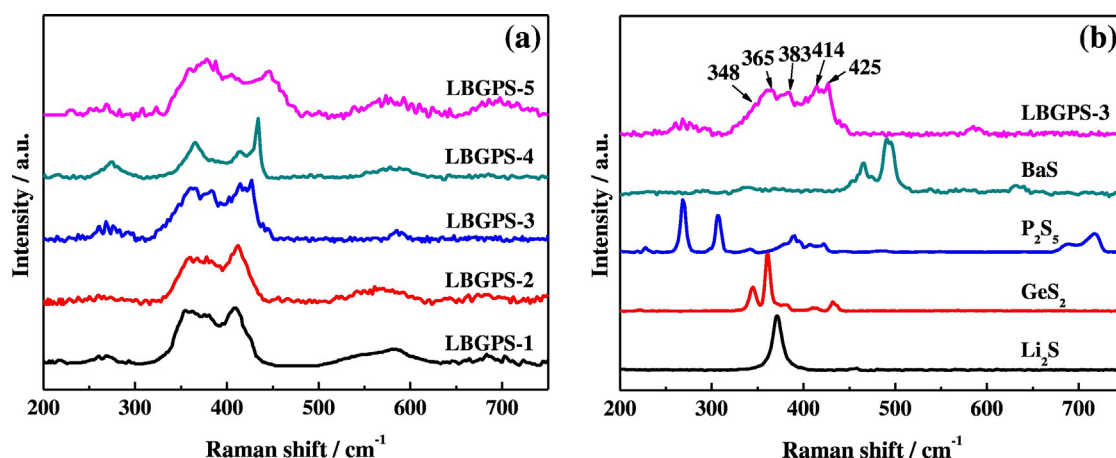


Fig. 2. Raman spectra of (a) Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ with different molar ratio of Li/Ba and (b) LBGPS-3 and the precursors.

sample LBGPS-3 has a relatively higher E_a value than that of LGPS, which is consistent with the ionic conductivity results.

To determine the lithium transference number in solid electrolyte, the chrono-amperometric measurement was conducted on a symmetric Li/LBGPS-3/Li cell. The measurement was carried out under a potential of 10 mV. Considering the influence of electrode–electrolyte interface resistance, the impedance measurement was performed on the same cell before and after the chrono-amperometric analysis. The impedance and polarization plots are shown in Fig. 4. The lithium transference number (t_{Li^+}) is calculated using the Bruce–Vincente–Evans equation [12]:

$$t_{Li^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} \quad (2)$$

where ΔV is the applied potential. I_0 and I_{ss} are the initial and steady state current, respectively, obtained from the polarization measurement. R_0 and R_{ss} are the interfacial resistance between electrode and electrolyte before and after polarization measurement. The impedance is characterized by the combination of two parts: the electrolyte resistance and the electrode–electrolyte interface resistance. To be specific, the intercept of the semicircle at the higher frequency is the electrolyte resistance, and the diameter of the semicircle contributes to the electrode–electrolyte interface resistance. An equivalent circuit was used to describe the impedance spectra in the form of $R_e(R_{int}Q_{int})$, where R_e is the electrolyte resistance, R_{int} and Q_{int} are the interface resistance and constant phase element, respectively. The derived parameters are $I_0 = 29.2 \mu A$, $I_{ss} = 25.8 \mu A$, $R_0 = 145.3 \Omega$, $R_{ss} = 190.7 \Omega$ as determined from the impedance and polarization profiles. The calculated lithium transference number is 0.99 for LBGPS-3. Compared with the conventional organic electrolytes, such as 1 M LiPF₆ in EC/DMC with a t_{Li^+} of

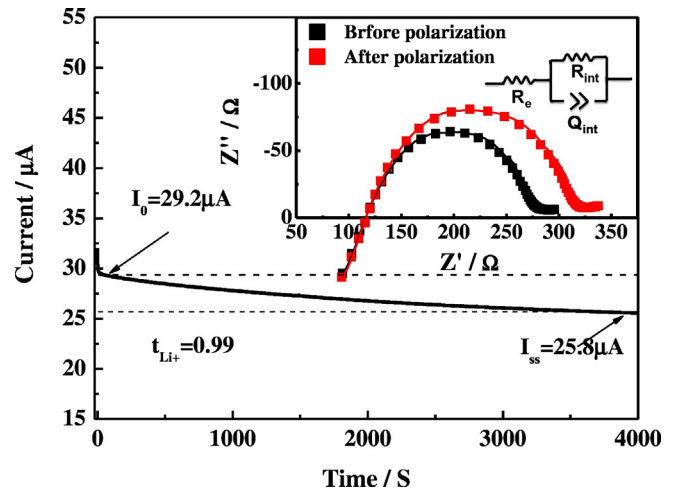


Fig. 4. Chrono-amperometric profile of a symmetric Li/LBGPS-3/Li cell under an applied potential of 10 mV. The inset shows the impedance plots performed before and after the polarization test.

0.21 and LiTFSI in DMC (1/20 = solvent/salt) with a t_{Li^+} of 0.49 [17], the synthesized material has a merit of enhanced cell rate capability by decreasing the concentration polarization.

Cyclic Voltammetry (CV) was carried out to evaluate the electrochemical stability toward the metallic lithium. A Li/LBGPS-3/stainless steel plate cell was assembled for the CV measurement, where stainless steel plate and Li foil worked as counter and reference electrode, respectively. The CV profile is shown in Fig. 5, the cathodic peak can be attributed to the lithium deposition $Li^+ = Li + e^-$, and the anodic peak is associated with the lithium stripping process $Li = Li^+ + e^-$. No obvious other peaks are found in the scanning range of -0.5 – 6 V due to the decomposition of solid electrolyte. However, different phenomena, such as sharp or broaden peaks, were observed in the traditional organic electrolytes [18]. Compared to the 5 V electrochemical window of LGPS reported by Kanno et al. [7], LBGPS-3 widens the electrochemical window toward high voltage by about 1.0 V. The improved structure via the doping Ba²⁺ in LGPS, which shields the coulomb repulsion among S²⁻, is favorable for the excellent electrochemical stability. This would enable the using high voltage cathode materials in Li-ion batteries, providing a facial strategy to increase the energy density of Li-ion batteries.

Fig. 6 displays the galvanostatic deposition–stripping spectra of LGPS and LBGPS-3 on a symmetric Li/SE/Li cell. With the increasing of charge

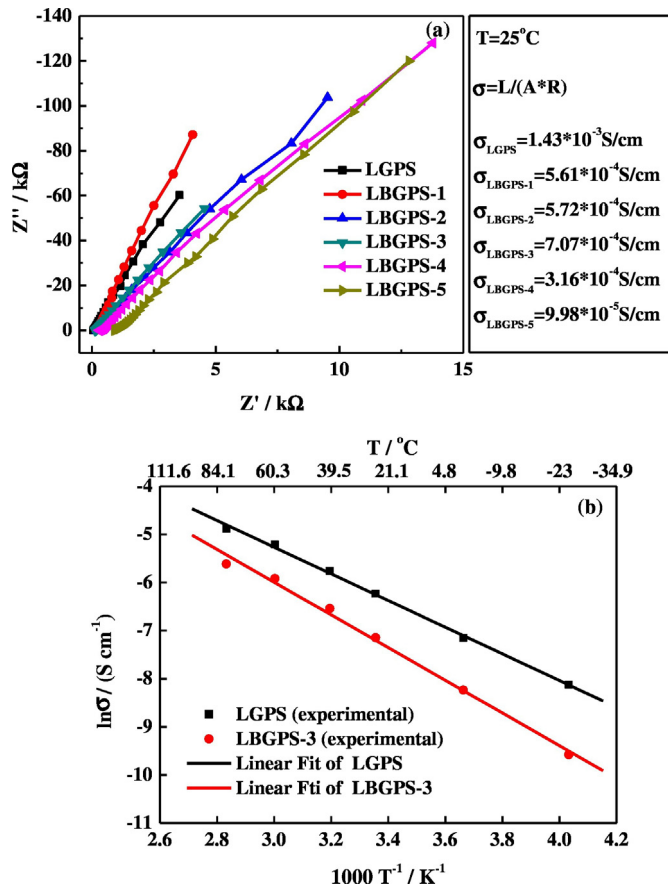


Fig. 3. (a) XPS survey spectra of CFO-ns and CFO-ns/rGO. (b), (c) and (d) High-resolution XPS spectra of O1s, Co 2p and Fe 2p for CFO-ns and CFO-ns/rGO.

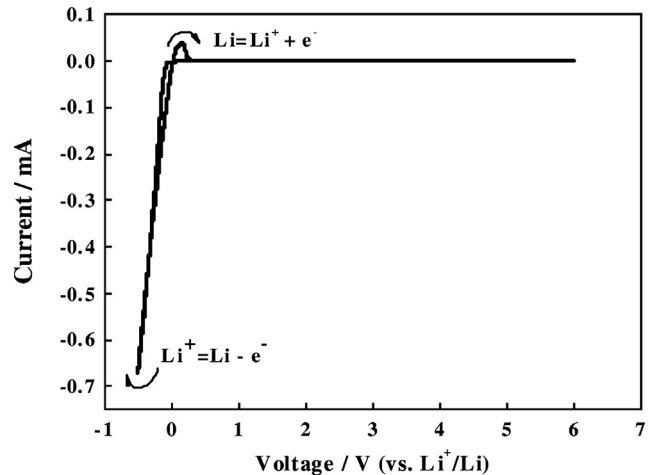


Fig. 5. Cyclic voltammetry of Li/LBGPS-3/stainless steel cell at a voltage between -0.5 and 6 V with a scan rate of 1 mV s^{-1} .

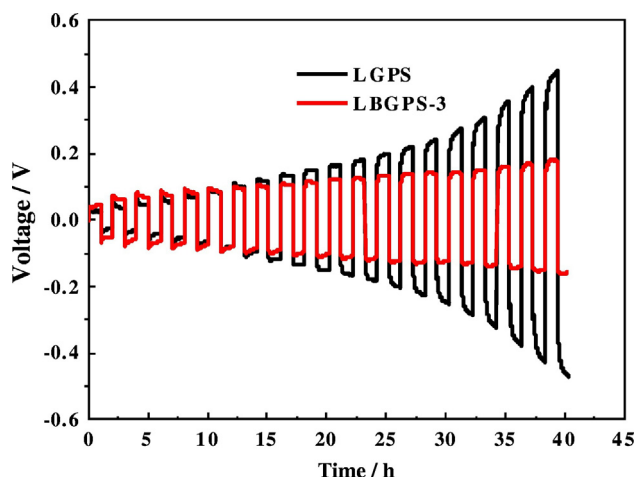


Fig. 6. Galvanostatic lithium deposition-stripping profiles of a symmetric Li/SE/Li (SE = LGPS, LBGPS-3) cell carried out at room temperature. Current: 0.1 mA.

and discharge cycle numbers, the polarization of Li/LGPS/Li cell is increased remarkably. While for the Li/LBGPS-3/Li cell, the polarization is significantly reduced and the cell demonstrates a considerable cycling stability. This suggests that the structural stability of LGPS is enhanced by the strong coulomb interaction between Ba^{2+} and S^{2-} as evidenced by the blue shift of the main peaks in Raman spectra (Fig. 2a). The improved structural stability propels the electrochemical stability of LGPS against the lithium electrode.

4. Conclusion

In this work, a facile method is proposed to improve the structural and electrochemical stability of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ by the divalent doping of Ba^{2+} in $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$. Doping content of Ba shows a significant impact on the ionic conductivity of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$. Among the Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (i.e. $\text{Li}_{10-2x}\text{Ba}_x\text{GeP}_2\text{S}_{12}$, $x = 0$ to 0.5) samples, $\text{Li}_{9.4}\text{Ba}_{0.3}\text{GeP}_2\text{S}_{12}$ exhibits the highest ionic conductivity of $7.07 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature. The wide electrochemical window and low polarization reveal that doping of Ba in $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ can improve the structural and electrochemical stability of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ effectively. The strong coulomb interaction between Ba^{2+} and S^{2-}

and screening the coulomb repulsion among S ions can account for the structural stability improvement. The high lithium ion transference number and excellent electrochemical stability render the Ba-substituted $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ a potential solid electrolyte candidate for all solid-state batteries with high energy density and safety. This work demonstrates that doping divalent element in $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ is a promising strategy for developing solid electrolyte with high structural stability.

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